

Project Proposal: Explainable Machine Learning System for Real-Time Financial Fraud Detection

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Introduction

Financial fraud in online payment systems continues to grow in scale and complexity, creating significant challenges for automated detection systems. While modern machine learning models can identify suspicious transactions with high accuracy, many operate as "black boxes," offering limited insight into how decisions are made. This lack of transparency can reduce trust among analysts and slow down decision-making in high-risk financial environments.

This project proposes the development of an explainable, real-time machine learning framework for financial fraud detection using the Cifer-Fraud-Detection-Dataset-AF. By combining streaming data technologies with interpretable prediction models, the system aims to provide both accurate fraud classification and clear, human-readable explanations for each alert.

Primary Computer Science Areas of Focus

This project focuses on Machine Learning, Data Engineering, and Distributed Systems. The core technical emphasis is on supervised learning and anomaly detection models for fraud classification, combined with explainable AI (XAI) methods to interpret model behavior. The system will leverage distributed architecture to handle high-frequency transaction streams.

Literature Review

1. Adaptive Ensemble Frameworks

Jin and Zhang (2025) addressed the limitations of single-model detection by implementing a Stacking ensemble that integrates diverse learners like SVM, Neural Networks, and Gradient Boosting Trees. A key technical contribution was their dynamic weight adjustment mechanism, which prioritizes the most effective model for specific transaction patterns. This approach yielded a 95% accuracy rate, demonstrating that weighted multi-model systems provide superior stability in fluctuating financial environments.

2. Scalable Ingestion and Processing

To handle the "velocity" problem of big data, Liu et al. (2025) proposed a distributed architecture using Apache Kafka and Flink for streaming, supported by Apache Spark for processing. Utilizing cloud services like AWS S3 and RDS, they achieved over 99% accuracy with Logistic Regression and Random Forest. Their research proves that real-time fraud prevention depends as much on the robustness of the data engineering pipeline as it does on the underlying machine learning logic.

3. Model Benchmarking and Class Imbalance

Alketbi (2025) conducted a high-dimensional comparison using the IEEE-CIS dataset, employing SMOTE to handle extreme class imbalances.

Random Forest was the most effective, reaching a 99.23% F1-score.

Deep Neural Networks (DNN) and Gradient Boosting maintained high precision, while Logistic Regression functioned as a fast but less accurate baseline.

Research Questions

To evaluate the effectiveness and scalability of the proposed system within a big data context, this project will address the following two research questions:

RQ1 (Scalability & Velocity): How can a distributed processing architecture maintain sub-second transaction scoring latency while managing the high "Velocity" and "Volume" of big data financial streams?

RQ2 (Explainability & Performance): To what extent can local explanation methods (such as SHAP and LIME) provide consistent, human-readable justifications for fraud alerts without compromising the system's real-time throughput requirements?

Key Performance Indicators (KPIs)

- **Model Accuracy and Reliability:** Achieve high precision, recall, and F1-score in detecting fraudulent transactions.
- **Explanation Fidelity:** Ensure that XAI-generated explanations (SHAP/LIME) accurately represent the model's decision-making logic.
- **System Latency:** Maintain low-latency processing from transaction ingestion to alert generation with attached explanations.

- **Concept Drift Detection:** Successfully identify and report shifts in data distribution that impact model performance over time.

Data Sources

We will utilize the **Cifer-Fraud-Detection-Dataset-AF** (available via Hugging Face), a comprehensive 21-million-row synthetic dataset modeled after real-world financial logs.

Column Features: The dataset includes step (temporal unit), type (transaction category), amount, oldbalanceOrig, newbalanceOrig, nameorig, namedest, isFraud, isFlaggedFraud, oldbalanceDest, and newbalanceDest.

Target Labels: The isFraud column serves as the ground truth, while isFlaggedFraud provides a baseline for traditional rule-based detection.

Why this dataset?

Its accounting-style features (sender/receiver balance shifts) are ideal for SHAP and LIME explanations, as they allow the model to "explain" fraud through logical balance discrepancies.

Tech Stack

Machine Learning: Python, Scikit-learn, XGBoost, Random Forest, SHAP, and LIME for explainability.

Data Engineering & Streaming: Apache Kafka for simulating real-time transaction flows.

Backend & API: FastAPI for serving predictions and explanations.

Monitoring: Custom dashboard for visualizing concept drift and model performance metrics.

Project Description

The proposed system will follow a pipeline-oriented architecture:

1. **Data Ingestion:** A Kafka producer will stream records from the Cifer dataset to simulate live transactions.
2. **Preprocessing:** Real-time feature engineering to handle categorical encoding (type) and calculate balance deltas.

3. **Fraud Classification:** An ensemble of XGBoost and Random Forest models will classify incoming transactions.
4. **Explainability Layer:** For every flagged transaction, the system will generate a SHAPbased explanation highlighting the specific features (e.g., unusually high amount relative to balance) that triggered the alert.
5. **Drift Analysis:** The system will monitor for "concept drift," identifying when new fraud patterns emerge that differ from the training data.

Conclusion

By integrating the Cifer-Fraud-Detection-Dataset-AF with explainable AI techniques, this project will demonstrate how financial institutions can move beyond black-box models. The system enables human analysts to understand the reasoning behind automated fraud alerts, fostering trust and enabling more effective responses in high-stakes environments.

References: -

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